That's where industry groups like ETSI come into play. Over 130 of the world's leading network operators have recently joined together to form an ETSI Industry Specification Group (ISG) for NFV. Literally dozens of groups — some open source and others more traditional standards organisations — are putting together pieces of the (large) puzzle needed to make NFV a reality. And operators (large and small) are engaging in proofs-of-concept exercises and evaluating the business case for what is surely the industry's largest transformation in decades.

OPNFV – An open platform to accelerate the NFV transformation

Open Platform for NFV (OPNFV) is an open source project that provides network operators an open source reference platform to validate multi-vendor, interoperable NFV solutions.

OPNFV facilitates the development and evolution of NFV components across various open source ecosystems. Through system level integration, deployment and testing, OPNFV creates a reference NFV platform to accelerate the transformation of enterprise and service provider networks. OPNFV will work closely with ETSI's NFV ISG, among other standards bodies, to drive the consistent implementation of an open and standard NFV reference platform.

The initial scope of OPNFV provides NFV infrastructure (NFVI), virtualised infrastructure management (VIM), and APIs to other NFV elements, which together form the basic infrastructure required for virtualised network functions (VNFs) and the management and network orchestration (MANO) components. Increasingly, standards are being drafted in conjunction with major open source projects. OPNFV will work with many of these projects to coordinate the continuous integration and testing of NFV solutions.

Participation is open to anyone, whether you are an employee of a member company or just passionate about network transformation. Some of the major open source projects for which collaboration with OPNFV is important are:

- *Virtual infrastructure management:* OpenStack, Apache CloudStack (*https://github.com/apache/cloudstack)*
- *Network controller and virtualisation infrastructure:* OpenDaylight (*https://github.com/opendaylight)*
- *Virtualisation and hypervisors:* KVM, Xen, libvirt and LXC (*https://git.kernel.org/pub/scm/virt/kvm/kvm.git)*
- *Virtual forwarder:* Open vSwitch (OVS) and Linux bridge
- *Data-plane interfaces and acceleration:* Data plane development kit (DPDK), Open data plane (ODP), etc. (*http://dpdk.org/dev)*

Figure 5: Open NFV (Source: *http://www.telecoms.com/wp-content/blogs.dir/1/files/2014/09/OPNFV.jpg*)

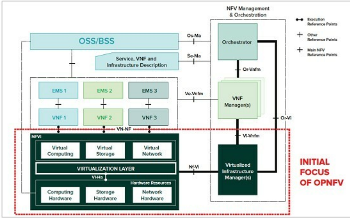

Figure 6: Open NFV focus today (Source: *https://www.sdxcentral.com/wp-content/uploads/2014/09/opnfv-nfv-open-source-network-functions-virtualization-diagram.jpg*)

Open Source MANO

Open Source Mano is an ETSI-hosted initiative to develop an open source NFV management and orchestration (MANO) software stack aligned with ETSI NFV.

ETSI's Open Source Mano (OSM) group is developing an open source NFV management and orchestration stack using well established open source tools and working procedures. The activity is closely aligned with the evolution of ETSI NFV and will provide a regularly updated reference implementation of NFV MANO. OSM aims at enabling an ecosystem of NFV solution vendors to rapidly and cost-effectively deliver solutions to their users.

Anybody who is interested can contribute here: *https://github.com/nfvlabs/openmano*

Open vSwitch

Open vSwitch (OVS) is a production quality, multi-layer virtual switch licensed under the open source Apache 2.0 licence. It is designed to enable massive network automation through programmatic extension, while still supporting standard management interfaces and protocols (e.g., NetFlow, sFlow, IPFIX, RSPAN, CLI, LACP, 802.1ag). In addition to exposing standard control and visibility interfaces to the virtual networking layer, it is designed to support distribution across multiple physical servers. Open vSwitch supports multiple Linux-based virtualisation technologies including Xen/XenServer, KVM and VirtualBox.

The bulk of the code is written in platform-independent C and is easily ported to other environments. The latest release of Open vSwitch has the following features:

Figure 7: Open Source MANO (Source: *https://osm.etsi.org/images/OSM-logo.png*)